MSc Financial Engineering Pergam Project Final Presentation

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Agenda

Project Overview

Reference paper. Wood, Roberts & Zohren (2022), *Slow Momentum with Fast Reversion: A Trading Strategy Using Deep Learning and Changepoint Detection*.

Research Question

Can a Deep Momentum Network (DMN) trained on the Sharpe ratio, augmented with Gaussian-Process Changepoint Detection (CPD), generate consistent out-of-sample alpha on European equities in a long-only, cost-aware deployment context?

- **Paper setting**: equity/commodity futures, bonds, FX, 1995–2020, long-short position, no transaction costs.
- **Our setting**: STOXX 600 single-stock equities, 2006–2026, long-only, realistic 25 bps costs when entering or closing a position.

Data and Universe Construction

Universe. STOXX 600 constituents, January 2006 – May 2026.

- **Survivorship-bias control.** Point-in-time constituent lists applied year-by-year: only tickers actually in the index that calendar year are retained.
- **Look-ahead bias controls.**
  - Returns shifted by -1 day: position at t is evaluated on return r_{t+1}.
  - 5-day forward-fill cap; year-boundary returns ($>$5 calendar days between observations) set to NaN.
  - EWMA volatility computed with `shift(1)` so same-day return is excluded.
- **Features**: normalised returns at 5 horizons ($h \in \{1, 21, 63, 126, 252\}$ days), 3 vol-normalised MACD signals (Baz *et al.* 2015), sector-relative return, GP-CPD severity ν_t and location γ_t.

Statistic	**Value**
Trading days	5,307
Unique tickers	1,314 (incl. delisted)
tickers/year	600
Stock-day observations	$\approx$1.6M

Changepoint Detection: Method Comparison

Motivation. Classical momentum signals react too slowly around regime turning points.

Denoising choice. CPD is run on *idiosyncratic* returns, not raw returns:

$$r^{(i)}_{\text{rel}}(t) = r^{(i)}(t) - r_{\text{benchmark}}(t) \qquad \text{(equal-weight benchmark)}$$

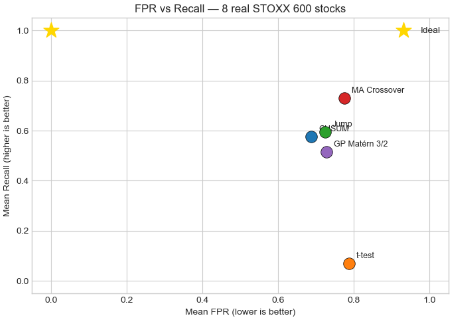
Removes market-wide co-movement to detect stock-specific breaks. On ASML: annualised vol 32% $\rightarrow$ 26% ($-$19% noise).

Method	Mean FPR $\downarrow$	Mean Recall $\uparrow$	Mean Delay $\downarrow$
CUSUM (Brownian)	59%	47%	1.9d
Jump process	71%	61%	2.0d
Rolling t-test	74%	7%	7.6d
MA crossover	65%	50%	2.5d
GP Matérn 3/2	**74%**	**50%**	**0.1d**

Why GP despite comparable raw metrics?

Naive detectors output a binary 0/1 flag – a step function the LSTM cannot learn by gradient. The GP outputs a continuous score $\nu \in [0, 1]$, calibrated as a Bayes-factor approximation, differentiable and directly usable as an LSTM input.

Changepoint Detection: FPR vs Recall trade-off



Interpretation (each method across 8 STOXX 600 stocks; ideal detector sits top-left):

- t-test collapses to near-zero recall – eliminated.
- Jump: best recall but highest FPR (noisiest).
- CUSUM: best balance among naive methods.
- GP: comparable on macro metric, wins on delay (0.1d) + continuous score.

The GP is chosen not for raw FPR/Recall, but because it is the only method producing a continuous, differentiable score usable by the LSTM.

Changepoint Detection: Scores & Scoring Limitation

The GP outputs **two continuous scores** per stock per day, both fed to the LSTM:

Severity:

$$\nu = 1 - \frac{1}{1 + \exp\big(-(\mathrm{NLML}_{\mathcal{M}_0} - \mathrm{NLML}_{\mathcal{M}_1})\big)} \to 1 \text{ when two regimes fit far better than one}$$

Location:

$$\gamma = \frac{c - (t - l)}{l} = \frac{c}{l - 1} \to 0 = \text{break long ago}, \ 1 = \text{break just now}$$

What the Matérn 3/2 kernel captures: three rupture types in a single fit

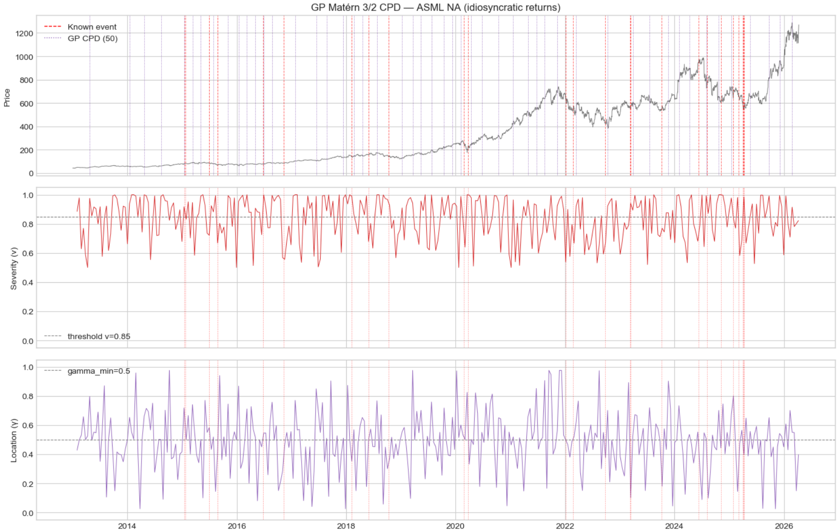
λ (input scale) $\to$ correlation-length change $\quad$ σ_h (output scale) $\to$ mean-reversion change
σ_n (noise) $\to$ volatility change

Empirically: ν spikes at COVID / Brexit / SVB on relative returns, with delay $\approx$ 0d – fastest of all methods.

Open limitation: idiosyncratic shocks

FPR/Recall metrics score only macro events, where all stocks move together. The genuinely interesting breaks are idiosyncratic (earnings, M&A) and cannot be scored without event data. The CPD evaluation is a partial validation – the true test is the out-of-sample Sharpe of the DMN + CPD backtest.

Changepoint Detection: GP Matérn 3/2 + CP kernel scores



LSTM Deep Momentum Network

Architecture.

- Input: sequence of 63 days $\times$ (8 or 10) features per stock.
- Single-layer LSTM (hidden size 20) $\rightarrow$ dense layer $\rightarrow$ tanh (long-short) or sigmoid (long-only).
- Output: position $X_t^{(i)} \in (-1, 1)$ or $(0, 1)$.

Loss function. Negative annualised Sharpe ratio of the vol-scaled strategy:

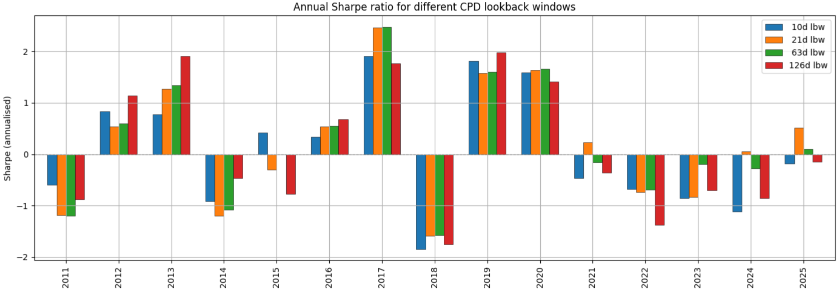
$$R_{t+1}^{(i)} = X_t^{(i)} \frac{\sigma_{\text{tgt}}}{\sigma_t^{(i)}} r_{t+1}^{(i)} \qquad \mathcal{L} = -\frac{\sqrt{252}\,\bar{R}}{\hat{\sigma}_R}$$

Quality-weighted transaction cost extension. The cost term weights each trade by whether the realised return confirms the position change:

$$\text{cost}_t = c\,\sigma_{\text{tgt}} \left| \Delta\left(\frac{X_t}{\sigma_t}\right) \right| \underbrace{\sigma\left(-\operatorname{sgn}\left(\Delta \frac{X_t}{\sigma_t}\right) \cdot \operatorname{sgn}(r_{t+1})\right)}_{\text{quality weight}}$$

A confirmed trade (direction matches outcome) is charged less; an unconfirmed trade pays the full cost. This reduces to the paper's flat-cost formula (Eq. C1) when the quality weight $\equiv 1$.

CPD Lookback Window Sensitivity



lbw $\in \{10d, 21d, 63d, 126d\}$ produce similar out-of-sample Sharpe profiles, with agreement in sign in (almost) every year, suggesting CPD timescale is not a critical hyperparameter on single-stock equities.

Note that lbw $= 252$ could not be evaluated under the expanding fold structure: fold 0's 5-year training window (2006–2010) is shorter than the combined CPD warm-up period (252 days) and LSTM sequence length (63 days), leaving no valid training sequences.

Walk-Forward Validation Design

Expanding folds (primary)

Fold	Train Period	N.Years	Test Period	N.Years
0	01/01/2006–31/12/2010	5	01/01/2011–31/12/2015	5
1	01/01/2006–31/12/2015	10	01/01/2016–31/12/2020	5
2	01/01/2006–31/12/2020	15	01/01/2021–31/12/2025	5

Rolling folds (robustness)

Fold	Train Period	N.Years	Test Period	N.Years
0	01/01/2006–31/12/2016	10	01/01/2017–31/12/2019	3
1	01/01/2010–31/12/2019	10	01/01/2020–31/12/2022	3
2	01/01/2013–31/12/2022	10	01/01/2023–31/12/2025	3

Each fold is trained from scratch (fresh random seed). Continuity between folds on the equity curve reflects concatenated OOS predictions, not a single model covering the full period.

Note that the LightGBM is trained with a different walk-forward design.

Why LightGBM? The Same Signals with More Data Support

Research question

Can the same causal signal families produce a tradable long-only strategy *without* recurrent sequence modelling?

Why test a lower-complexity design?

- **Observed LSTM ratio:** 6,202 sequences / 2,581 parameters $= 2.4\times$.
- **Reference range:** $10\times$–$100\times$ examples per parameter. This gap indicates a clear overfitting risk.
- **LightGBM changes the balance:** 2.8M stock-day rows, 1,311 tickers, and 300 shallow depth-4 trees.

20 causal features

Feature family	#
Normalised returns (1–252d)	5
MACD (8/24, 16/48, 32/96)	3
Realised volatility	1
Lag-1 copies	9
Lagged GP-CPD (ν, γ)	2
Total	**20**

Same causal inputs, different inductive bias. Because stock-day rows are correlated and tree complexity is not directly comparable with neural-network parameter counts, the claim is a *more favourable data-to-complexity balance*, not an equivalent LightGBM ratio.

LightGBM Predicts Returns, Not Positions

Gradient boosting, MSE objective

1. Start from a constant return forecast.
2. Fit a shallow tree to the residuals.
3. Add it with learning rate $\eta = 0.05$: $\widehat{r}_{t+1}^{(m)} = \widehat{r}_{t+1}^{(m-1)} + \eta\, f_m(x_t)$.
4. Repeat up to 300 trees; regularisation caps complexity.

Complexity controls

Parameter	Value
Max depth	4
Max leaves	15
Min leaf size	20
Row subsample	0.8
Feature subsample	0.8
L_1 / L_2	0.1 / 0.1

Key distinction

The model only minimises error on r_{t+1}. Turning forecasts into portfolio positions is a *separate, validation-only* calibration step (next slides).

Annual Walk-Forward: Every Test Year Stays Unseen

1. Train

Fit LightGBM on the expanding history (MSE).

2. Validate

Pick α^* on the last 10% of the window — net Sharpe, 25 bps.

3. Test

Freeze model & α^*; predict the next year once.

$$\underbrace{2006:2010}_{\text{train/val}} \rightarrow \boxed{2011}\ ;\ \underbrace{2006:2011}_{\text{train/val}} \rightarrow \boxed{2012}\ ;\ \cdots\ ;\ \underbrace{2006:2025}_{\text{train/val}} \rightarrow \boxed{2026}$$

Out-of-sample record

Sixteen annual folds are concatenated from January 2011 to May 2026. No test observation is used for fitting or calibration.

Validation signal

Mean fold IC $= 0.0245$; positive in 15 of 16 folds. The signal is small but directionally consistent. This is descriptive evidence, not a formal significance test.

Scores Become Cross-Sectional Long-Only Exposures

Each day, raw scores are standardised across stocks, then passed through a sigmoid:

$$s_t^{(i)} = \frac{\widehat{r}_{t+1}^{(i)} - \overline{\widehat{r}}_{t+1}}{\mathrm{sd}_i\left(\widehat{r}_{t+1}\right)}, \qquad \widetilde{X}_t^{(i)} = \sigma\big(\alpha^* s_t^{(i)}\big) \in (0,1).$$

Why standardise by date?

- Daily return forecasts are tiny in absolute terms.
- The z-score turns them into a cross-sectional *ranking*.

What does α^* do?

- Sets how aggressively ranks become exposure.
- Search grid: 40 log-spaced values, $0.1 \leq \alpha \leq 10$.
- Chosen on time-ordered validation by net Sharpe after the 10-day EMA and 25 bps costs; then frozen for the test year.

Reading a position of 0.5

0.5 is neutral *relative to the signal*, but still a 50% long exposure — not market-neutral and not a flat position.

EMA Smoothing Halves Position Variability

Independent daily predictions make exposure jumpy. A 10-day half-life EMA produces implementable positions:

$$\overline{X}_t = \lambda \overline{X}_{t-1} + (1-\lambda)\widetilde{X}_t, \qquad \lambda = \exp\left(-\frac{\log 2}{10}\right) \approx 0.933.$$

Before smoothing

$$\mathrm{sd}(\widetilde{X}) = 0.0576$$

After the EMA

$$\mathrm{sd}(\overline{X}) = 0.0275$$

- Position variability drops by $\approx 52\%$.
- Lower day-to-day change $\Rightarrow$ less turnover and noise.
- The EMA changes *implementation*, not the return forecast.

GP-CPD Acts as a Risk Filter, Not a Forecast

The lagged GP-CPD severity $\nu_t \in [0, 1]$ pulls exposure back toward the neutral anchor when a regime break is detected:

$$X_t = 0.5 + (1 - \nu_t)\,(\overline{X}_t - 0.5).$$

Calm regime $(\nu_t \approx 0)$

- Smoothed position left almost unchanged.

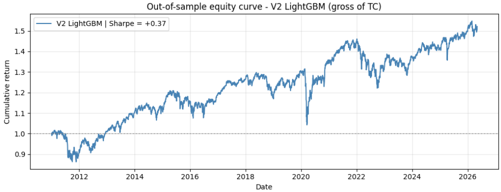
Break detected $(\nu_t \approx 1)$

- Exposure pulled toward 0.5 — never reversed.

Observed impact

The filter moves 28.1% of out-of-sample positions by more than 0.01 — a targeted risk guardrail. SHAP confirms CPD is not the main return driver.

Gross vs. Net: Different Questions, One Signal



Notebook equal-weighted LightGBM curve, **gross** of costs and before the common NB04 volatility framework.

Same strategy, three reporting scales

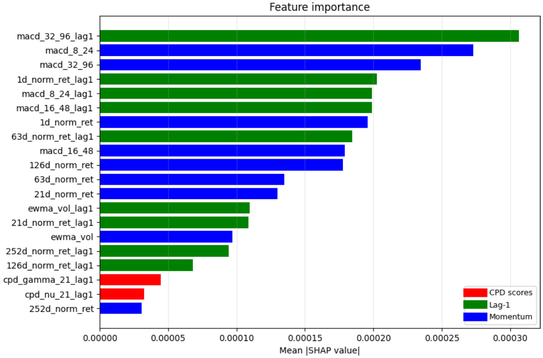
Basis	SR	Ret.	M
Raw gross (NB03)	.366	3.09%	−20.
Common net (NB04)	.268	1.09%	−9.
Net, 15% vol	.268	4.02%	−31.

Common net backtest: 25 bps costs, 60-day EWMA vol scaling, daily hit ratio 53.7%.

Interpretation

Costs cut Sharpe from 0.366 to 0.268. The 15%-vol rescaling only rescales return and drawdown — Sharpe is unchanged. The three rows are reporting scales, not competing estimates.

SHAP: The Signal Lives in Momentum, Not CPD



What the model uses

- MACD(32/96) lag-1 is the top driver.
- Short-horizon normalised returns follow.
- Both GP-CPD variables rank near the bottom.

Takeaway

- Predictive content is mostly momentum persistence; CPD adds risk control.

Interpretation boundary

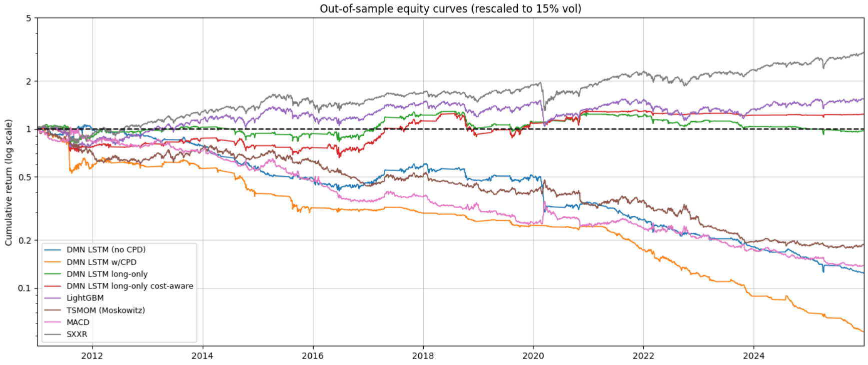
SHAP shows how the fitted model *allocates attribution*; it does not prove economic causality or guarantee future stability.

Full-Period Performance Table

Strategy	Return	Sharpe	MDD	Hit %	Avg Holding (d)
DMN LSTM (no CPD)	−15.72%	−1.048	−88.35%	43.1%	1.1
DMN LSTM w/ CPD	−28.43%	−1.896	−98.89%	37.6%	1.1
DMN LSTM long-only	+0.85%	+0.056	−28.67%	50.5%	1.6
DMN LSTM long-only + TC	+3.27%	+0.218	−35.58%	51.7%	2.5
LightGBM w/ CPD + TC	+4.02%	+0.268	−31.48%	53.70%	1.2
TSMOM (Moskowitz)	−9.30%	−0.620	−82.28%	51.1%	—
MACD	−11.92%	−0.795	−87.16%	50.8%	—
SXXR (cap-weighted)	+8.46%	+0.564	−33.45%	54.2%	—

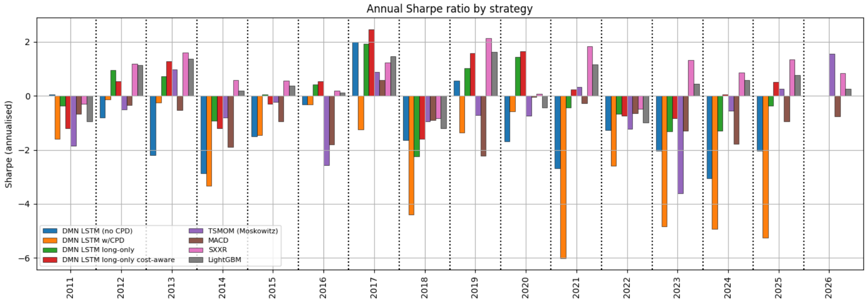
Net of 25 bps backtest costs applied uniformly to all strategies. Vol target $\sigma_{\text{tgt}} = 15\%$.

Equity Curves



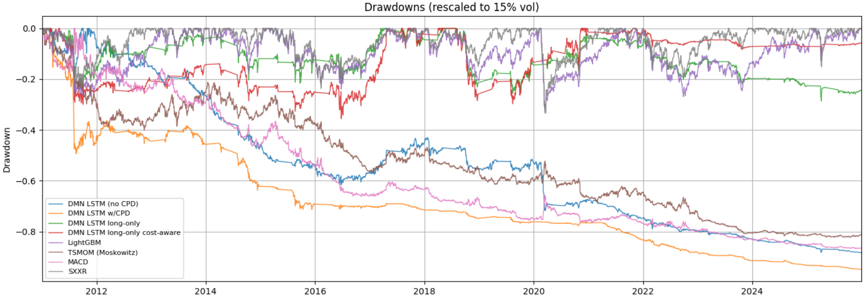
All strategies rescaled to 15% annualised volatility for risk-adjusted comparability. SXXR is the buy-and-hold cap-weighted benchmark; Moskowitz *et al.* (2012) strategy is calculated as $X_t = (1-w)\,\mathrm{sgn}(r_{t-252,t}) + w\,\mathrm{sgn}(r_{t-21,t})$; MACD is the sign of the average of three vol-normalised MACD signals with $(S, L) \in \{(8,24),(16,48),(32,96)\}$.

Annual Sharpe Ratios



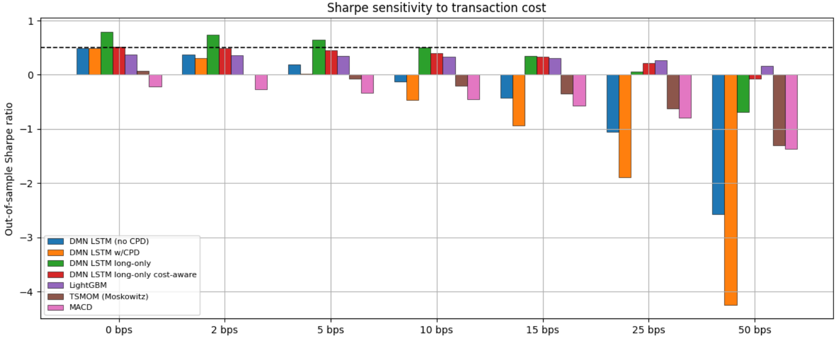
The year-by-year decomposition reveals strong regime-dependence. Both LSTM and LightGBM generate positive Sharpe in trending markets (2012, 2017, 2019, 2021) and struggle in mean-reverting or high-volatility regimes (2018, 2022).

Drawdown Analysis



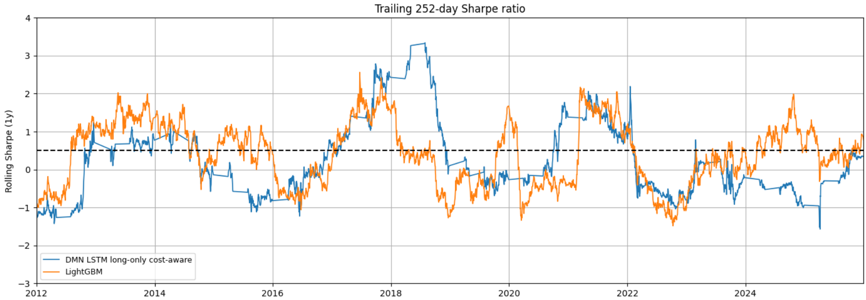
Drawdowns computed on vol-rescaled cumulative returns. The long-only DMN variants exhibit shallower drawdowns than the cap-weighted SXXR benchmark, consistent with the strategy's implicit momentum timing (reduced exposure after regime breaks).

Transaction Cost Sensitivity



The quality-weighted cost extension produces more graceful Sharpe degradation as costs increase, compared to the flat-cost baseline. LightGBM, with its built-in EMA position smoothing, also shows relatively robust cost sensitivity due to lower intrinsic turnover.

LSTM vs. LightGBM Sharpe Comparison



Both models agree in sign in most years, indicating that the momentum signal dominates over the choice of model architecture. The LSTM tends to generate larger Sharpe swings; LightGBM is more conservative.

Robustness Check: Expanding vs. Rolling Folds

Rolling folds (3y test windows) produce qualitatively consistent results, validating that the expanding fold findings are not an artefact of growing training sets.

Strategy	**Return**	**Sharpe**	**MDD**	**Hit %**	**Avg Holding (d)**
DMN LSTM (no CPD)	−30.88%	−2.058	−90.52%	38.6%	1.1
DMN LSTM w/ CPD	−40.84%	−2.723	−91.82%	33.1%	1.1
DMN LSTM long-only	+2.11%	+0.140	−29.97%	51.4%	1.5
DMN LSTM long-only + TC	+5.36%	+0.357	−27.45%	54.0%	1.9
LightGBM w/ CPD + TC	+4.02%	+0.268	−31.66%	53.70%	1.2
TSMOM (Moskowitz)	−9.30%	−0.620	−82.28%	51.1%	—
MACD	−11.92%	−0.795	−87.16%	50.8%	—
SXXR (cap-weighted)	+8.46%	+0.564	−33.45%	54.2%	—

Net of 25 bps backtest costs applied uniformly to all strategies. Vol target $\sigma_{\mathrm{tgt}} = 15\%$.

Robustness Check: GARCH(1,1) vs. EWMA

Strategy	Vol Model	Return	Sharpe	MDD	Calmar
LSTM	EWMA (60d)	+0.67%	+0.218	−7.83%	+0.085
LSTM	GARCH(1,1)	+0.54%	+0.188	−7.37%	+0.073
LightGBM	EWMA (60d)	+1.09%	+0.268	−9.61%	+0.114
LightGBM	GARCH(1,1)	+0.89%	+0.232	−8.10%	+0.110

Positions kept fixed; only the vol-scaling denominator changes. Sharpe ratios are largely stable across volatility estimators, confirming that the signal generation — not the scaling — drives performance. Calmar ratios differ slightly due to path-dependence in the compounded equity curve.

Conclusions

1. We replicated the Wood *et al.* (2022) pipeline on STOXX 600 European equities and adapted it for a long-only fund context with realistic t.c. **The paper's results do not transfer directly**: long-short DMNs produce catastrophic drawdowns on single-stock equities (>-80% at 15% vol).
2. The **long-only constraint** is the single most important structural modification: it aligns the strategy with the equity risk premium rather than fighting it.
3. Our **quality-weighted cost extension** provides a principled improvement over the paper's flat-cost Sharpe loss, reducing unnecessary turnover while preserving signal fidelity, and producing more graceful Sharpe degradation under increasing t.c.
4. The **GP-CPD module** provides limited but consistent marginal value on equities, acting mainly as a risk filter rather than a return driver. This is partly an evaluation-design issue: we validated CPD against *global macro* events, but the DMN consumes it *per stock* – idiosyncratic, single-name regime breaks are a different (and noisier) target than the macro shifts the method was tuned to detect.

Conclusions (cont'd)

5. **LightGBM** is a competitive, faster-to-train alternative, achieving similar Sharpe to the LSTM with no sequence modelling required.
6. The **SXXR buy-and-hold** remains a strong benchmark over 2011–2025, setting a high bar for active strategies in a period dominated by the equity risk premium and low rates.

Appendix : GP-CPD Mathematical Formulation

1. **Returns and standardisation**. Arithmetic return and per-window standardisation, $\mathcal{T} = \{t-l, \ldots, t\}$:

$$r_t^{(i)} = \frac{p_t^{(i)} - p_{t-1}^{(i)}}{p_{t-1}^{(i)}} \qquad \hat{r}_t^{(i)} = \frac{r_t^{(i)} - \mathbb{E}_{\mathcal{T}}[r_t^{(i)}]}{\sqrt{\mathrm{Var}_{\mathcal{T}}[r_t^{(i)}]}}$$

2. **GP prior**. $\hat{r}_t = f(t) + \epsilon_t$, $f \sim \mathcal{GP}(0, k_\xi)$, $\epsilon_t \sim \mathcal{N}(0, \sigma_n^2)$, with the Matérn 3/2 kernel ($\xi_M = (\lambda, \sigma_h, \sigma_n)$):

$$k(x, x') = \sigma_h^2 \left(1 + \frac{\sqrt{3}|x-x'|}{\lambda}\right) e^{-\sqrt{3}|x-x'|/\lambda}$$

3. **Marginal likelihood**. $p(\hat{\mathrm{r}} \mid \xi) = \mathcal{N}(0, V)$, $V = K + \sigma_n^2 I$; hyperparameters fit by minimising

$$nlml_\xi = \frac{1}{2}\hat{\mathrm{r}}^\top V^{-1}\hat{\mathrm{r}} + \frac{1}{2}\log|V| + \frac{l+1}{2}\log 2\pi \quad \text{(L-BFGS-B)}$$

4. **Changepoint kernel**. A hard region-switching kernel at c is discrete and expensive to optimise; instead we blend two Matérn 3/2 kernels k_{ξ_1}, k_{ξ_2} with a sigmoid $\sigma(x) = 1/(1 + e^{-s(x-c)})$:

$$k_{\xi_C}(x, x') = k_{\xi_1}(x, x')\,\sigma(x)\sigma(x') + k_{\xi_2}(x, x')\,\bar{\sigma}(x)\bar{\sigma}(x'), \qquad \xi_C = \{\xi_1, \xi_2, c, s, \sigma_n\}$$

$c \in (t-l, t)$ is the changepoint location; $s > 0$ the transition steepness (large $s \to$ a hard switch, recovering the discrete kernel).

5. **Severity and location**. Comparing $nlml_{\mathcal{M}_0}$ (single Matérn) against $nlml_{\mathcal{M}_1}$ (changepoint kernel):

$$\nu_t^{(i)} = \sigma\left(nlml_{\mathcal{M}_0} - nlml_{\mathcal{M}_1}\right) \qquad \gamma_t^{(i)} = \frac{c-(t-l)}{l}$$

Appendix: CPD Alternative Methods

Benchmarked in `notebooks/02_changepoint_detection.ipynb` against known macro events (tolerance ± 20 trading days), aggregated over 5 stocks.

- **CUSUM** (online, mean + variance). Two-sided cumulative sum on standardised returns z_t, plus a variance statistic:

$$S_t^+ = \max(0, S_{t-1}^+ + z_t - k), \quad V_t = \max(0, V_{t-1} + (z_t^2 - 1) - k_v)$$

  Flags when $\max(S_t^+, S_t^-) > h_\mu$ or $V_t > h_\sigma$; continuous score $\nu = \tanh(\max(S^+, S^-, V)/h)$.
- **BOCPD** (Bayesian online, Adams-MacKay 2007). Maintains a full posterior over run-length r_t under a Normal-Inverse-Gamma model; score is $P(r_t \leq 5 \mid r_{1:t})$ – a well-calibrated probability.

CUSUM actually outperforms GP on this binary-detection benchmark. Cross-method score correlation is low ($\rho \approx 0.12$–0.14), so the three continuous scores capture largely independent information rather than the same signal.

Appendix: Overfitting Diagnostics

- **Train vs. validation Sharpe gap (LSTM).** Early stopping (patience = 25 epochs) limits overfitting. Dropout (rate 0.3) provides regularisation. The train/val gap narrows monotonically across expanding folds as training data grows, consistent with a model that generalises rather than memorises.
- **LightGBM capacity.** Shallow trees (max depth 4, $\leq$15 leaves) with L1+L2 regularisation. The signal-to-noise ratio (IC $\approx$ 0.005) is the binding constraint, not model capacity.
- **Fold stability.** Sharpe variation across folds correlates with known macro regimes (2018 vol spike, 2020 COVID, 2022 rate hike cycle) rather than exhibiting random noise, suggesting regime-dependence rather than overfitting.

Appendix: Raw Full-Period Performance Table

Strategy	Return	Vol	Sharpe	MDD	Hit %	Avg Holding (d)
DMN LSTM (no CPD)	−1.13%	1.08%	−1.048	−13.57%	43.1%	1.1
DMN LSTM w/ CPD	−1.78%	0.94%	−1.896	−16.41%	37.6%	1.1
DMN LSTM long-only	+0.10%	1.84%	+0.056	−3.71%	50.5%	1.6
DMN LSTM long-only + TC	+0.67%	3.06%	+0.218	−7.83%	51.7%	2.5
LightGBM w/ CPD + TC	+1.09%	4.08%	+0.268	−20.75%	53.70 %	1.2
TSMOM (Moskowitz)	−2.44%	3.93%	−0.620	−34.32%	51.1%	—
MACD	−3.45%	4.34%	−0.795	−42.89%	50.8%	—
SXXR (cap-weighted)	+9.04%	16.03%	+0.564	−35.36%	54.2%	—

Net of 25 bps backtest costs applied uniformly to all strategies.